

Tag & Track Sprint 2

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1. Introduction

- Introductions - Tag and Track Team
 - Martin Hundrup - Team Lead and CS Major, Back End
 - Sarah Carley - CS Major, Front End
 - Shawn Will - CS Major, Database
- Objectives
 - Begin frontend and backend implementation
 - Provide some application functionality
 - Further research database implementation and show proof of concept

2. Sprint Objectives

- Create an interactive User Interface.
- Integrate a basic database.
- Provide QR scanning support.

These components will help us build a solid foundation to build the rest of the application off of.

3. Feature Implementation (Martin)

- Features: QR Scanner, QR Generator, Data Management, Debug Logger
 - Showcase: (in video)
 - Testing and Validation: Only informally

3. Feature Implementation (Shawn)

- Feature: Database
 - Showcase: N/A
 - Testing and Validation: We tested the database implementation by just checking to make sure all updates were properly saved to the device, and that the data stayed consistent between boots.

3. Feature Implementation (Sarah)

- Feature: UI Components
 - Showcase: Images attached of some of the components made.
 - Testing and Validation: These were tested by creating the components on the frontend and validated by ensuring that the components looked and worked as intended.



ID	Arctos ID	Name	Description	Status
0	ARC-000000	Salmon Specimen	Cleaned bone specimen	true
1	ARC-000001	Lion Claw	Preserved specimen stored in fluid jar	true
2	ARC-000002	Bear Femur	Egg sample in container	false
3	ARC-000003	Heron Egg	Fluid jar sample with tag	true
4	ARC-000004	Toad Sample	Full skeleton display	false
5	ARC-000005	Owl Wing	Tissue sample for genetics	true

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4.1. Figma Prototype

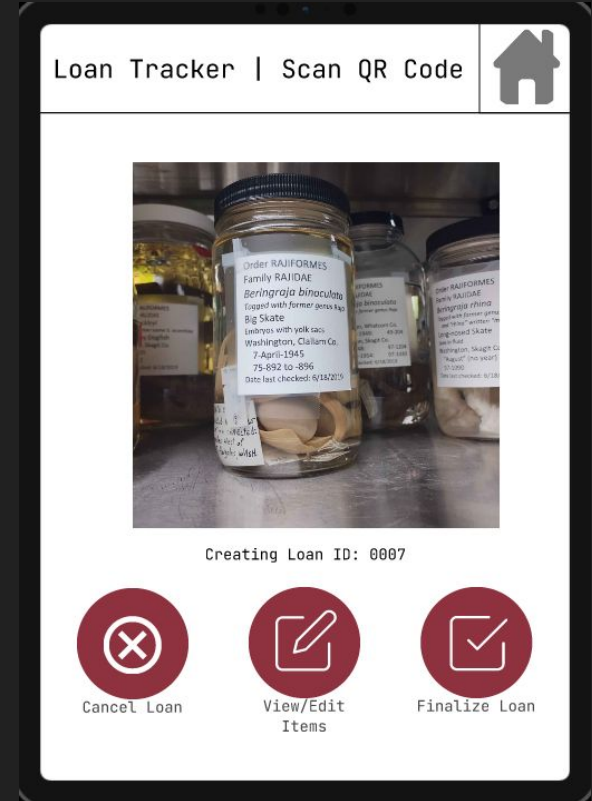
Prototype

4.2. Client Feedback on Figma Prototype

- Client was pleased with the prototype.
- The prototype was aligned with what the client wants from this project overall.
- We plan on basing much of our UI off of this prototype.

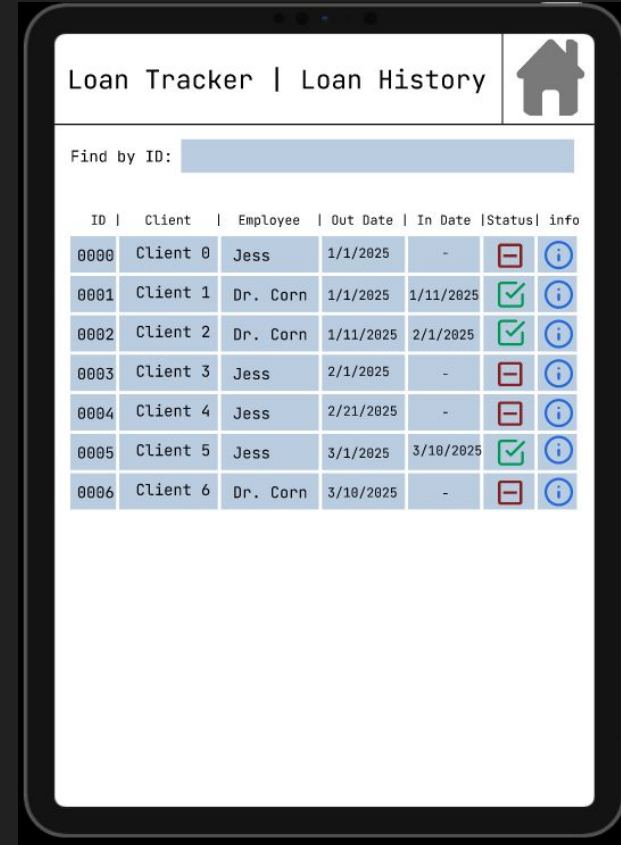
4.3. Figma Prototype Component Explanation

- QR scans for loan making
 - Designed to have a majority of the page be used for seeing the camera's view.
 - Contains options for loans such as cancelling, viewing, and finalizing that will map to the associated pages.



4.4. Figma Prototype Component Explanation

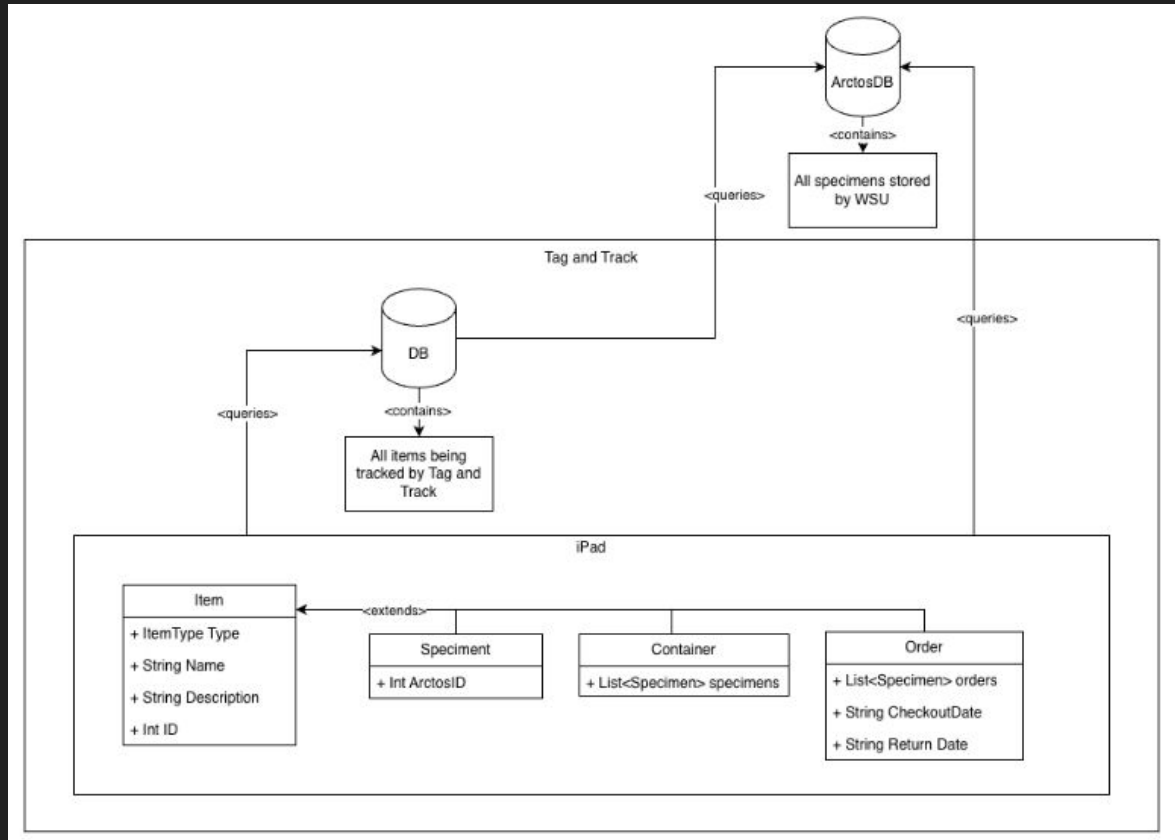
- Data page for loan history
 - Designed to have a list of previous loans.
 - General design of the data table able to be used across other pages that require a data table to be shown.
 - Will map to a page with loan information if loan viewed.



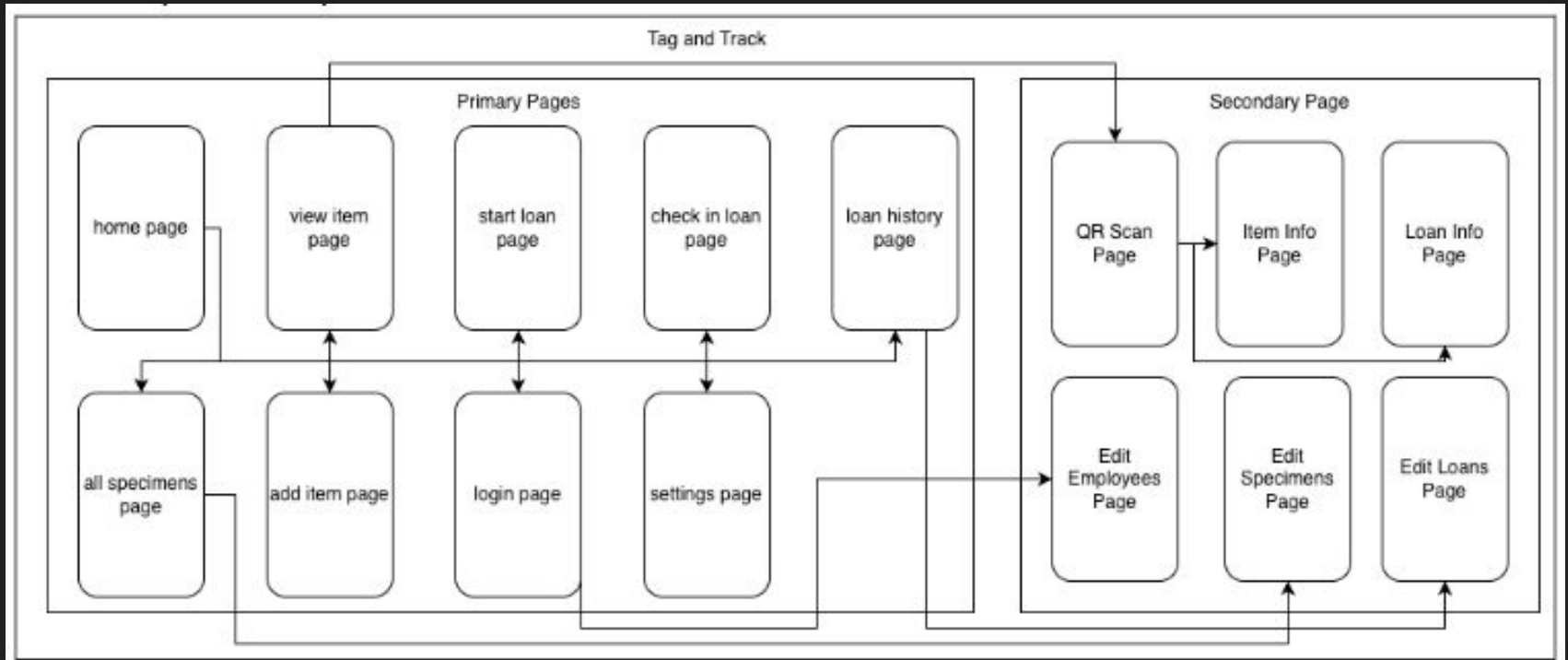
5. Kanban Overview and Contributions

Kanban Board

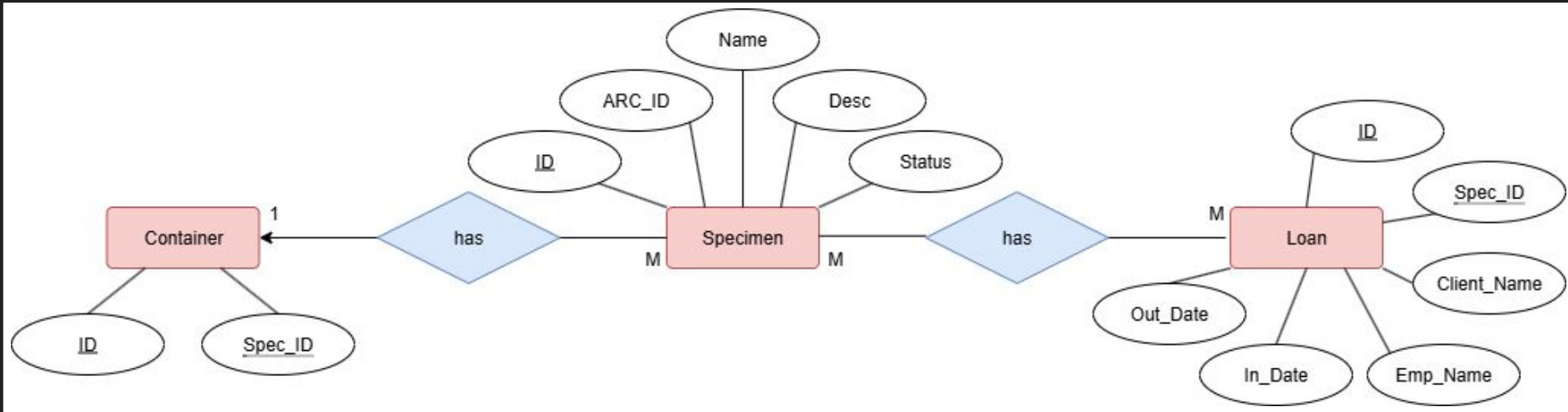
7.1.1 System Architecture



7.1.2 System Architecture



7.2.1. Database Design (ER Diagram)



7.2.2. Database Design (Structure/Relationships)

We will be implementing an API to connect to the Arctos Database, which will act as our external database. Since we do not control this database, I won't go into detail about what it contains. However, what we plan to do is implement our own internal database that roughly mirrors Arctos. Using SQLite, we will set up our own database to store the basic information needed for checking in and out specimens from the museum. This information includes the specimen ID and specimen name, with room to add any extra info if the museum staff want.

There will also be two other databases part of SQLite that will store all the check in and out data (also known as loan info), and the container info (which are the containers that hold multiple specimens). Both of these tables will have references to the specimen database as to save space.

7.2.3. Database Design (Key Tables)

Specimen	Loan	Container
int ID (PK)	int ID (PK)	int ID (PK)
string ARC_ID	int spec_ID (PK) → Specimen(ID)	int spec_ID (PK) → Specimen(ID)
string Name	string client_name	
string Desc	string emp_name	
bool Status	dateTime out_date	
	dateTime in_date	

7.2.4. Database Design (Optimization)

Our database is not going to be crazy large, so optimization isn't crazy important. However, this isn't to say we won't keep it in mind. When designing the database, we will look to create search functions that aren't too computationally heavy. If need be, indexing will be considered to help reduce search time.

7.3. User Interface Design

- Based on client feedback, our UI will be based closely on the Figma Prototype design.
- This sprint, the main UI focus was the creation of components that will likely be reused in different pages across the application. Examples of these components include:
 - A button template.
 - A data table template.
 - A text box template.



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7.4. Testing Plan

- As of now, we have yet to implement formal tests. However, we plan on doing the following:
 - Unit testing: creating tests tailored to specific methods to ensure their capability.
 - Testing with NUnit: we will likely use NUnit to perform our tests.
 - White box testing: We will use our knowledge from implementation to craft the related tests.
- Informal testing:
 - Visual validation: test frontend and backend changes by viewing them on the UI once implemented to test.
 - Debug validation: ensure that the outcomes of methods are as expected by watching variables during debugging while running the code.

8. Sprint Achievements and Challenges

Achievements

- UI and backend integration.
- Local database creation and usage.

Challenges

- Development in a somewhat unfamiliar framework (.NET Maui).
- Arctos database integration.

Improvements from Sprint 1

Due to Mac issues, we had no code in Sprint 1. This sprint shows a drastic improvement, as we were able to implement a frontend, a database, and QR scanning capabilities.

9. Next Step and Retrospective

Next steps:

- Work more with the database and connect more to the backend.
- Create/edit more pages with UI components.
- Implement testing.
- Make a MVP (minimum viable product) for the next sprint.

Retrospective:

What went well

- We have a strong base for the rest of our program.
- QR Code scanning.
- Ipad testability.

What went poorly

- Arctos API - we were unable to get the API key during the time of the sprint.

Thank you